Supplementary file 1 — GRRAS checklist

Guidelines for Reporting Reliability and Agreement Studies (Kottner et al., 2011)

Manuscript: *What can be measured from the broadcast? Validity of single-camera video for football-kick kinematics and swing-leg joint loads.* Maintained repository checklist; section numbers refer to the repository reading version. The published SportRxiv supplement is preserved separately in docs/preprint (DOI 10.51224/SportRxiv.1131).

#	GRRAS item	Where it is reported
1	Identify in title or abstract that interrater/intrarater reliability or agreement was investigated	Title ("validity"); abstract ("method-comparison study", agreement against motion capture)
2	Name and describe the diagnostic or measurement device of interest explicitly	§1 (broadcast-video pipeline); §2.3 (RTMW-x detector, MotionAGFormer-B lifter, scaling, alignment, contact)
3	Specify the subject population of interest	§1 (football players kicking in broadcast matches); §2.1 (laboratory participants as reference population and its limits)
4	Specify the rater population of interest (if applicable)	Not applicable: the method is automated (detector + lifter). The reference "rater" is optical motion capture (§2.1)
5	Describe what is already known about reliability and agreement and provide a rationale for the study	§1
6	Explain how the sample size was chosen; state the number of raters, subjects/objects and replicate observations	§2.7 (fixed by the public datasets; 83 participants); §2.1 (7 CMU kicks); §2.3 (5 views × 5 sizes × 2 frame rates; 3 noise seeds for CMU, 1 for MoVi)
7	Describe the sampling method	§2.1 (all seven CMU kick trials; all MoVi participants with a usable kick, after interpolating gaps of up to 0.25 s and excluding kicks with longer gaps; all 200 3DSP shots)
8	Describe the measurement process	§2.2 (features, loads, windows); §2.3 (virtual camera, error models, lifting); §2.4 (real video, digital twin); §2.5 (remedies)
9	State whether measurements were conducted independently	§2.3: scale, one rigid alignment, kicking-leg identity and known contact use reference information in the main conditions; these are optimistic benchmark conditions. The adapted lifter uses unseen participants and held-out broadcast residuals (§2.5). The posture-surrogate transfer model is fitted on all MoVi laboratory observations and its video agreement is not held out (§3.5).
10	Describe the statistical analysis	§2.7 (bias, MAE, limits of agreement, ICC(2,1), calibration slope, bootstrap CIs, exploratory operational criteria, TOST, custom paired sign-flip waveform permutation tests); §2.8 (hypothetical information-fraction sensitivity; no KLI-specific event estimate or season forecast)

#	GRRAS item	Where it is reported
11	State the actual number of raters and subjects included and the number of replicate observations	§2.1 and Table 1; §3.4 (162 usable recordings from 82 participants; two participants with one usable camera); Tables 2–4 (n per analysis)
12	Describe the sample characteristics of raters and subjects	§2.1 (CMU: 2 participants; MoVi: 83 participants, 55 female, non-footballers, air kicks)
13	Report estimates of reliability and agreement including measures of statistical uncertainty	§3.3–3.7; Tables 2–4 (bias, MAE with 95 % CI, 95 % LoA, ICC with 95 % CI); Supplementary Figures
14	Discuss the practical relevance of results	§4.1–4.4
15	Provide detailed results if possible (e.g., online)	Supplementary file 2 (complete statistical outputs); code and aggregated results in the public repository